



A New Weighting Scheme for Diminishing the Effect of Extreme Values in Regional Drought Analysis

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Abstract

Climate warming has increased the risk of recurrent drought hazards. Previous research has indicated potential associations between climate warming and extreme climate events. Due to short distance spatial variability in meteorological variables, the prevalence of extreme values may result in biased inferences or reduce the scope of regional drought analysis. Therefore, precise inferences of spatiotemporal characteristics of regional drought are challenging. In this study, we provide a new weighting scheme to diminish the effect of extreme values for the regional aggregation of precipitation data. We used K – Components Gaussian Mixture Models (K–CGMM) based on the standardization procedure to achieve maximum computational accuracy. Consequently, the article proposes a new regional drought index – the Bias Diminished Weighted Regional Drought Index (BDWRDI). We assessed the performance of BDWRDI by calculating the Standardized Precipitation Index (SPI) using data from six meteorological stations in the Northern area of Pakistan. We found that the proposed weighting scheme can greatly diminish the effect of extreme values during the spatiotemporal aggregation of precipitation data. In addition, the strong positive correlation of BDWRDI with individual SPIs endorses the proposed weighting scheme and the proposed drought indicator for regional drought analysis.

Keywords Climate warming · Extreme values · Drought monitoring · Correlation

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1 Introduction

Due to climate change and global warming, frequent drought occurs in many parts of the world. Compared to other hazards, the effects of drought are more prolonged and devastating. Drought adversely impacts agriculture (Azadi et al. 2018), ecosystem (Gampe et al. 2021), energy (Byers et al. 2020), health (Sugg et al. 2020), and several other aspects of life. The severe impact of drought in underdeveloped countries can go beyond economic losses, causing severe starvation and possibly human mortality (Bahinipati 2020; Magalhães and Magalhães 2019; Salvador et al. 2020). Unlike other natural hazards, drought is a complex climatic process (Mishra and Singh 2011). The main factors of drought occurrences are insufficient water in the ground over an extended period, high evapotranspiration, low humidity, high temperature, and high wind speed. Therefore, its accurate monitoring is a challenging task (Balti et al. 2020).

Accurate monitoring and assessment of drought help to reduce its severe impacts (Adedeji et al. 2020). In recent decades, continuous assessment and its estimation have received considerable attention. Recent research proposes several drought monitoring tools to measure and analyze various drought characteristics. In this perspective, Standardized Drought Indices (SDIs) type monitoring tools are widely and frequently used to quantify and assess drought hazards. Some most commonly used drought indices include Palmer Drought Severity Index (PDSI) (Palmer, 1965), Standardized Precipitation Index (SPI) (McKee et al. 1993), Reconnaissance Drought Index (Tsakiris et al., 2017) Standardized Precipitation Evapotranspiration Index (SPEI) (Vicente et al., 2010), Joint Deficit Index (JDI) (Kao and Govindaraju 2010), Multivariate Standardized Drought Index (MSDI) (Hao et al., 2013), Seasonally Transient Weighted Multi-Scaler Standardized Index (STWMSDI) (Ali et al. 2021a, b).

The aforementioned drought indices require different climatic parameters for drought characterization. Further, the applications of most of these indices are regionally specified. Svoboda and Fuchs (2016) have provided a comprehensive list of drought indices and their corresponding parameters. Parallel to drought monitoring tools, several drought forecasting algorithms have been developed for future drought assessment. Some most commonly forecasting algorithms include Autoregressive Integrated Moving Average (ARIMA), Wavelet Neural Network (WNN), Support Vector Machine (SVM) (Zhang et al. 2020), and the Wavelet-based hybrid ANN-ARIMA models (Khan et al. 2020), etc.

The structural differences in the numerical values of SDIs at various rain gauge stations scattered in a specific region always exist. Therefore, the primary interest of drought management is to see the picture of drought at the regional level. However, the procedure of SDIs depends on the information from a single meteorological station. Therefore, a comprehensive regional drought monitoring index is essential for regional drought management. In recent research, some efforts have been made to develop regional drought monitoring. For example, Ali et al. (2020) have proposed a steady-state probability-based regional drought index – The Spatially Weighted Accumulated Drought Index (SWADI). Raza et al. (2021) have proposed a Seasonally Blended Regionally Integrated Drought Index (SBRIDI) using Bayesian network theory to capture the seasonal and regional variability in the homogenous regions.

As a consequence of global warming, spatiotemporal changes in climatic variables and their patterns lead to the recurrent occurrence of extreme precipitation (Gregersen et

al. 2013). The hydrologic cycle will intensify with global warming, likely increasing the intensity of extreme precipitation and the risk of extreme events such as drought and flood (Tabar, 2020). So extreme precipitation is one of the potential outcomes of global warming. In several studies, extreme precipitation events are projected to become more frequent with future global warming (Allan and Soden 2008; Chevuturi et al. 2018; Papalexiou and Zaroug, 2018; Giorgi et al. 2019; Yaduvanshi and New, 2021).

The secondary goal for regional drought analysis is accurately accumulating the time series of rain gauge station observations. In usual data aggregation techniques such as model averaging, extreme events demolish the representativeness characteristics of data. For example, extreme values greatly affect the arithmetic mean and cannot be used for skewed data. Since precipitation distribution is non-normal, addressing the extreme precipitation observations in regional drought monitoring is essential for accurate and representative regional analysis.

Hence, it is more appropriate to accumulate spatial precipitation data under some medially weighted strategy. Under extreme values, this study aims to: (1) develop a new weighting scheme for accumulating rain gauge precipitation station observations in a particular region, and (2) develop a regional drought monitoring indicator for effective drought management.

The main contributions of this paper are three folded: First, the article provides a new weighted regional drought index – the BDWRDI. Second, the article estimates the SPI of the individual station, simple averaged based SPI and BDWRDI for the whole region under study. Third, to check the efficiency and appropriateness of BDWRDI, the research assesses the correlation of BDWRDI and model-averaged based SPI with individual station-based SPI.

2 The proposed weighting scheme for regional data aggregation

This section describes the steps involved in the development of BDWRDI. Figure 1 shows the flowchart of the BDWRDI. The primary purpose of the proposed weighting scheme is to reduce the impact of extreme values and outliers in the aggregation of precipitation data to avoid under-representative regional inferences. This section comprises the rationale and mathematical explanations of the proposed weighting scheme for aggregating the time series data of precipitation of various meteorological stations located in a specific region. The distinguishing characteristic of the proposed weighting scheme is to give more weights to those values whose incidence is more concerning homogeneity among them. On the other hand, divergent values will receive lower weights. This makes the rationale for reducing the impact of extreme values and outliers in the aggregation process. Consequently, the amalgamated data under the proposed weights will be more representative at the regional level.

The stepwise execution of the proposed weighting scheme is as follows:

Let $F \in (X_1, X_2, \dots, X_K)$ be a multivariate time series data of precipitation recorded at various meteorological stations in a specific region. Where Y_i be the time series data of median computed from each row from F . Here, the main interest of the weighting scheme is to implicate the absolute difference between each observed value in various stations of F and median (i.e., $d_i = |y - x_i|$) in such a way, those meteorological stations with a short distance from the median value receive large weights. Accordingly, short weights should be

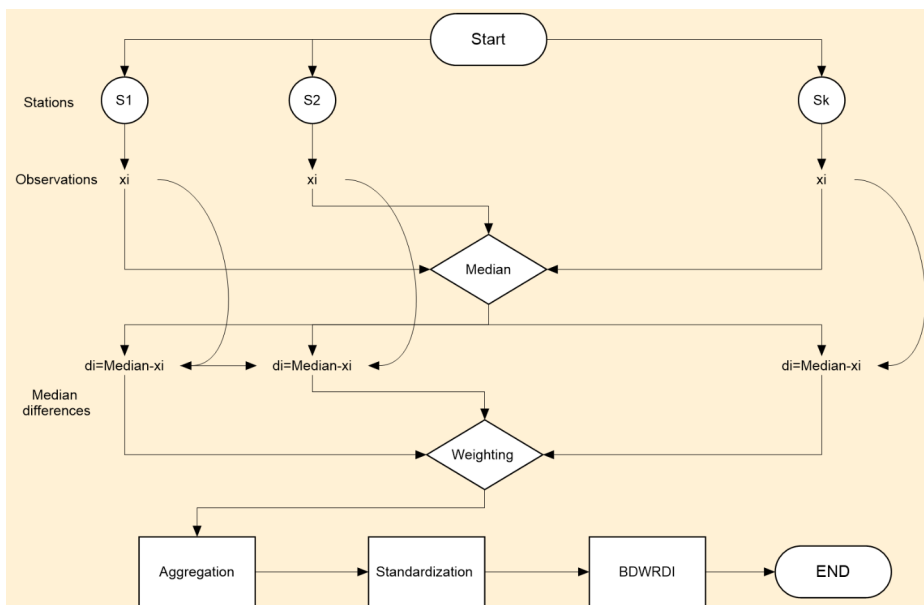


Fig. 1 Flowchart of the proposed index

assigned to those meteorological stations with large differences from the median value. To accomplish this task, we put forward the subsequent steps:

a) Step 1. Exponentials of the absolute differences.

Let $d_i = |y - x_i|$ be the differences in the observed value of precipitation of i^{th} meteorological station data from the median. Then the first step suggests taking the exponent of i^{th} differences. Mathematically, Eq. 1 describes the first step of the proposed weighting scheme.

$$h_i = e^{d_i} \quad (1)$$

Here, the objective of the above equation is to maximize the differences.

b) Step 2. Approximation of weights.

To assign low weight to larger deviated values and high weight to small deviated or close values with median, the second step suggests the following transformation.

$$v_i = 1 - \frac{h_i}{u} \quad (2)$$

In the above equation, $u = \sum_{i=1}^k h_i$.

c) Step 3. Standardizing weights.

The third step standardizes v_i in such a way that $\sum_{i=1}^k v_i = k$. The expression of standardization is as follows:

$$w_i = \frac{v_i}{\sum_{i=1}^k v_i} \quad (3)$$

Subject to the condition that $\sum_{i=1}^k w_i = 1$. Here, k represents the total number of meteorological stations in a region under consideration.

d) Step 4. Data aggregation under the proposed weights.

By the rationale of weighted average, this step suggests aggregating precipitation data of various meteorological stations. Mathematically,

$$p_c = \sum_{i=1}^k x_i w_i \quad (4)$$

e) Step 5. Spatiotemporal data aggregation.

Finally, step 5 suggests iterating steps 1 to 4 for all the time series data. The following expressions describe the mathematical form of the resultant weights for each station at each time.

$$w_{it} = \frac{v_{it}}{\sum_{i=1}^k v_{it}} \quad (5)$$

$$p_{ct} = \sum_{i=1}^k x_{it} w_{it} \quad (6)$$

In the above equation, p_{ct} represents a single univariate regionally aggregated data under the proposed weighting scheme.

2.1 Derivation of the Bias diminished Weighted Regional Drought Index (BDWRDI)

After aggregating spatiotemporal data under the above weighting scheme, this research integrates K-CGMM to derive a new drought index – the BDWRDI drought index. Under the rationale of the weighting scheme, we hypothesize that the BDWRDI is a more accurate and efficient regional drought indicator. The main aim of this index is to diminish the influence of extreme values and outliers. After the aggregation and application of the proposed weighting scheme on precipitation time series, the following is the main step involved in deriving BDWRDI.

2.1.1 The choice of Probability Model: the K-Component Gaussian Mixture Model (K-CGMM)

One of the essential phases in calculating SDI is fitting an appropriate probability distribution on the time series data of different climatic variables. Substantial precautions are required to choose the appropriate probability model. On the other hand, the poor fitness of the probability model reduces the scope of inferences (i.e., accuracy and reliability in the estimates). In past research, several univariate probability distributions have been suggested for modeling precipitation data for the computation of SDI. However, many authors have used gamma distribution to compute the SPI drought index (McKee et al. 1993; Bong et al., 2020). In the recent decade, Stagge et al. (2015) and Ali et al., (2020) have suggested various probability models. The appropriate probability function can be selected using the ‘*fitdistrplus*’ and ‘*Propagate*’ R packages (Delignette-Muller et al., 2013; Spiess et al., 2018).

However, it is quite possible to have more than one peak of the data, or the data may follow the multimodal distribution. In this case, univariate probability models are not suitable for accurate inferences. On the other hand, multi-model distribution or a mixture of probability functions can enhance computational accuracy. From this perspective, Ali et al. (2021a, b) have suggested using K-CGMM for modeling precipitation time series.

In light of the above argument, this research employs K-CGMM to obtain SPIs (for individual stations), SPI (regionally averaged under equal weights), and BDWRDI. Readers are referred to see the for more details on the standardization process.

After standardization, the resultant values of BDWRDI can be classified unto various various drought classes.

3 Application

The application of this research is based on the six meteorological stations of Gilgit-Baltistan, also called the Northern region of Pakistan (see Fig. 2). The selected region has significant importance in the water resource management system of the country. The region is located on the second-highest plateau of the world. The selected meteorological stations along with latitude and longitude are Astor (35.36 N, 74.84E), Bunji (35.64 N, 74.63E), Chilas (35.43 N, 74.08E), Gilgit (35.92 N, 74.30E), Gupis (36.22 N, 73.44E), and Skardu (35.30 N, 75.61E). Due to climate change and global warming, water resources have lost their reservoir at an unprecedented rate. Consequently, drought becomes hazardous and prominent in the region. In this study, time-series data of monthly accumulated precipitation accumulated from (1971-to 2016) were used to assess the implications of the proposed model for more accurate regional drought monitoring. The data has been collected from Pakistan Metrological Department Islamabad (<https://www.pmd.gov.pk/en/>). The summary statistics of the data, including arithmetic mean, standard deviation, and kurtosis of precipitation, are presented in Table 1.

The following sections describe the SPI as a threshold measure and Pearson correlation as comparative statistics.

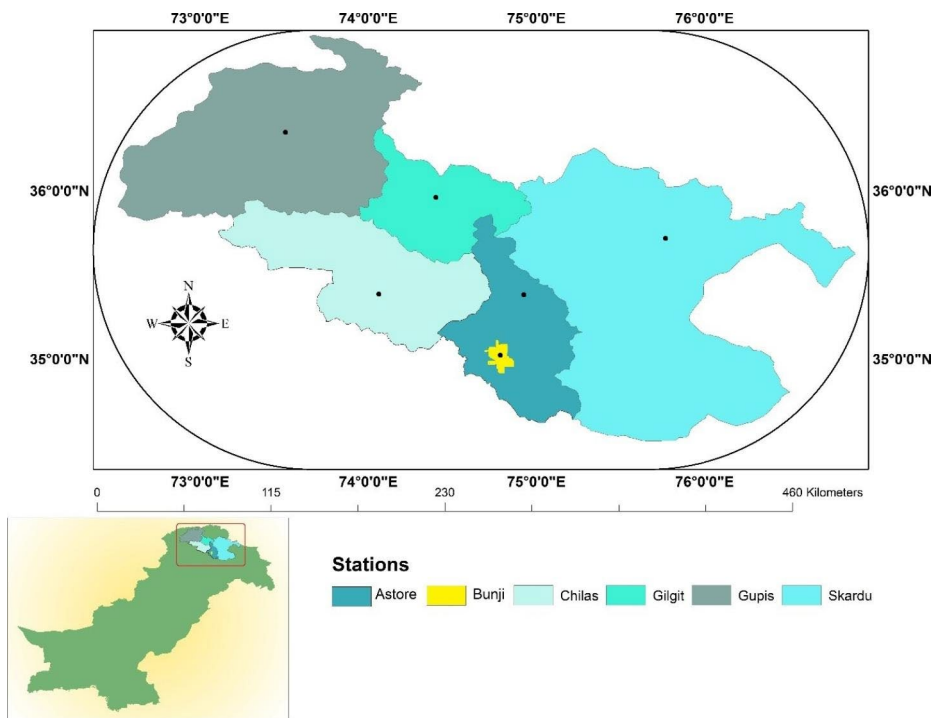


Fig. 2 Geographical locations of the study region

3.1 Standardize Precipitation Index (SPI)

In this study, the consistency and efficiency of the proposed index are compared with SPI under Pearson correlation. SPI is the most commonly used drought monitoring index among all the standardized drought indices. In past research, several authors have compared the performance of the newly proposed drought index with SPI (Huang et al. 2020; Kamruzzaman et al. 2019). The computation procedure of SPI is simple and easy. The stepwise procedure of SPI computation is as follows: The first step determines the time window or time scale of the monthly accumulative time series of precipitation. In this study, we have calculated SPI as well as the proposed drought index for -1 , -3 , -6 , -12 , -24 , and -48 time scales. In the second step, the precipitation time series is modeled using an appropriate probability distribution. We modeled the SPI and the proposed drought index using the K-CGMM. The use of K-CGMM is due to Ali et al. (2021a, b).

3.2 Quality measure and comparative assessment

The Pearson correlation coefficient (r) is one of the most commonly used statistical measures for assessing the linear relationships between two variables. The range of the correlation coefficient is from -1 to $+1$. A positive value very near to $+1$ indicates a strong positive correlation. In contrast, values near -1 show a strong negative or inverse relationship. Moreover, $r = 0$ shows the absence of a linear relationship between variables. Numer-

Table 1 Summary statistics of the selected stations

Months	Descriptive Statistics	Precipitation (mm)					
		Astor	Bunji	Chilas	Gilgit	Gupis	Skardu
January	Avg.	37.55	6.13	10.74	3.66	7.60	29.23
	Std.	37.88	8.02	16.07	6.31	11.60	29.46
	Kurtosis	1.610	6.35	5.26	10.14	3.57	1.05
February	Avg.	45.84	9.79	16.17	7.15	13.11	29.67
	Std.	35.75	14.65	15.43	8.90	26.98	25.02
	Kurtosis	0.290	8.94	4.96	4.36	23.66	3.30
March	Avg.	70.32	12.19	28.17	11.31	16.82	40.10
	Std.	44.32	12.04	26.54	11.03	17.95	36.31
	Kurtosis	1.230	7.42	1.79	2.58	1.45	1.43
April	Avg.	77.90	25.65	35.63	24.84	43.09	29.38
	Std.	52.25	25.79	33.53	28.19	75.04	30.93
	Kurtosis	0.150	1.50	1.39	3.76	11.14	3.21
May	Avg.	61.15	26.36	30.97	25.45	27.08	26.02
	Std.	51.10	26.98	35.89	24.06	30.54	31.05
	Kurtosis	-0.19	2.72	3.94	-0.10	0.81	3.62
June	Avg.	25.95	11.02	10.99	10.91	18.74	9.45
	Std.	20.74	10.33	13.96	9.19	28.10	9.82
	Kurtosis	5.36	0.16	5.60	4.83	5.19	0.25
July	Avg.	25.70	17.56	12.65	14.95	16.18	10.45
	Std.	19.94	16.15	14.46	12.66	18.17	9.44
	Kurtosis	2.28	2.65	5.35	1.49	4.61	4.40
August	Avg.	26.55	22.87	17.69	17.21	26.40	14.83
	Std.	22.13	23.98	18.55	18.60	29.05	14.07
	Kurtosis	5.81	2.61	7.77	4.46	8.29	3.11
September	Avg.	27.19	15.26	10.47	11.25	13.55	13.14
	Std.	33.06	19.83	20.84	12.65	21.94	23.07
	Kurtosis	8.31	2.31	24.63	6.25	12.93	13.23
October	Avg.	21.04	6.91	6.70	6.35	8.82	6.83
	Std.	29.78	19.28	22.68	16.31	17.17	17.75
	Kurtosis	15.10	35.43	36.35	30.21	19.67	29.40
November	Avg.	19.48	3.26	6.18	3.31	2.80	6.46
	Std.	25.70	8.53	12.88	5.82	6.51	12.67
	Kurtosis	3.33	18.18	6.25	6.39	8.14	10.84
December	Avg.	29.79	5.51	10.61	5.12	4.77	16.29
	Std.	38.71	9.91	18.80	8.66	9.67	20.81
	Kurtosis	5.10	13.90	5.51	5.21	12.65	4.29

ous studies reported the use of correlation analysis to compare various measurements. More specifically, correlation analysis is commonly used for the comparative evaluation of drought indicators. Zhang et al. (2018) compared the modified PDSI with the SPI and SPEI using a correlation coefficient.

Liu et al. (2021) investigated the correlation between the SPEI and Net Primary Productivity (NPP) drought index. Recently, Wei et al. (2021) comprehensively compared various drought indices and events with different time scales using the Pearson correlation coefficient. The Pearson correlation coefficient is one of the oldest and most applicable measure-

Table 2 Averaged weights for various months against each station

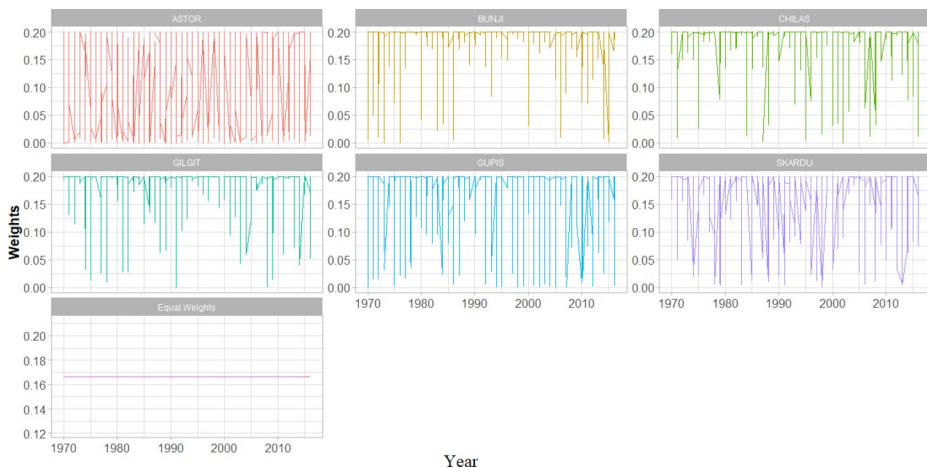
Months	Astor	Bunji	Chilas	Gilgit	Gupis	Skardu
Jan.	0.089034	0.194879	0.184585	0.195979	0.184034	0.151491
Feb.	0.065904	0.197891	0.190851	0.199211	0.19103	0.155117
Mar.	0.043943	0.194304	0.196477	0.195621	0.195204	0.174451
Apr.	0.048513	0.1996	0.198162	0.195313	0.174198	0.184213
May.	0.095834	0.188821	0.189584	0.183721	0.154494	0.187545
Jun.	0.114194	0.183794	0.188906	0.180657	0.14924	0.183204
Jul.	0.13903	0.165991	0.177181	0.182102	0.168577	0.167117
Aug.	0.149551	0.177111	0.179898	0.184094	0.137943	0.171411
Sep.	0.113464	0.188598	0.182604	0.184934	0.157606	0.172804
Oct.	0.082936	0.187755	0.184417	0.184679	0.180838	0.179394
Nov.	0.082936	0.187755	0.184417	0.184679	0.180838	0.179394
Dec.	0.082621	0.196464	0.186149	0.192364	0.177906	0.164487

ments widely used to assess linear relationships between random variables in ecology and water resource sciences. In the current study, we have used the correlation coefficient to evaluate the candidate-ship of the proposed drought index for regional drought monitoring.

4 Results and discussion

4.1 Estimation of weights and regional aggregation of precipitation data

This section presents the results associated with the estimation of weights under the proposed weighting scheme for regional data aggregation. Under the proposed weighting scheme, Table 2 shows the averaged weights in all the months against each station. One of the aims of this study was to assign unequal weight to each month at each station. By the rationale of the proposed weighting scheme, the inequality of weights adjusts the impact of extreme values in regional aggregated data. One source of inequality in weights is related

**Fig. 3** Time series data of weights under proposed weighting scheme and equal weighting scheme

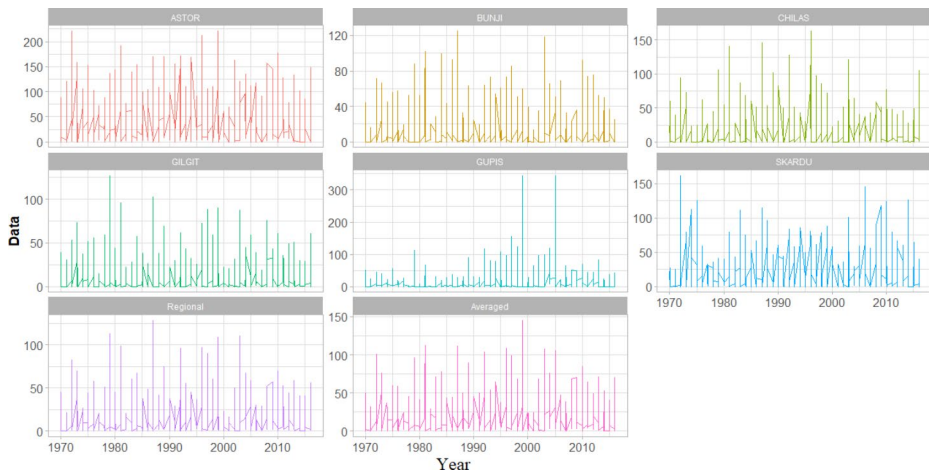


Fig. 4 Time series data of precipitation at individual stations, averaged and regionally proposed

to the temporal behavior of precipitation. At the same time, the additional variation might result from a varying altitude of meteorological stations. The results of this study indicate that there is significant variation in weights both spatially and temporally. It can be observed from the data in Table 2 that the overall average weight of January (0.089034) for Astor station is comparatively low than other stations.

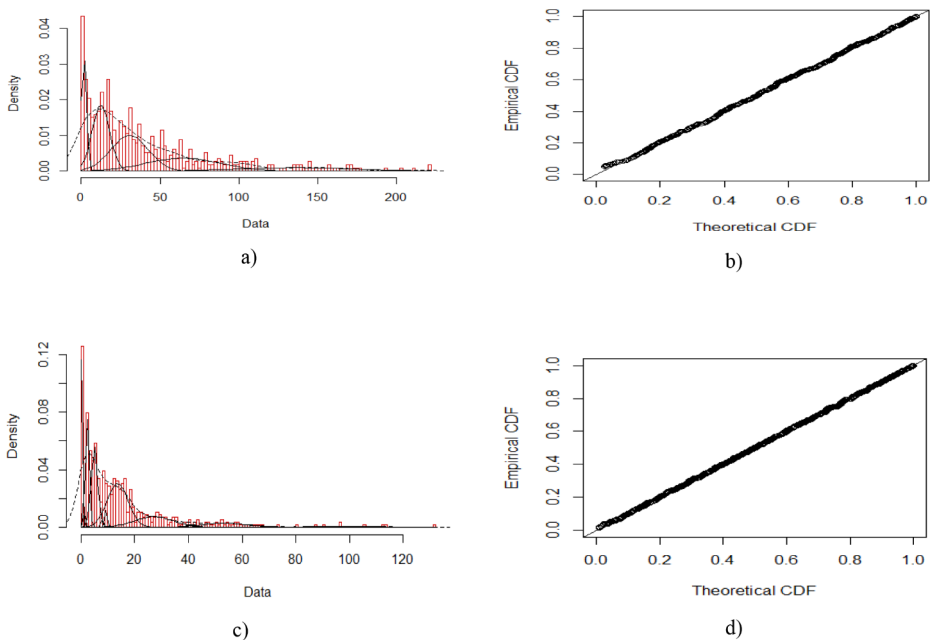


Fig. 5 Density plots and scatter plot of K-CGMD: (a) Density plot for SPI-1 for Astor, (b) Scatter plot of Empirical CDF vs. Theoretical CDF of SPI-1 for Astor, (c) Density plot for BDWRDI, (d) Scatter plot of Empirical CDF vs. Theoretical CDF of BDWRDI-1 for Astor

In contrast, the weights of Bunji, Chilas, Gilgit, and Gupis are homogenous. From Table 2, the spatiotemporal variations in weights are evident. Looking at Fig. 3, it's apparent that the weights in each month for each station significantly diverge. Consequently, in the presence of extreme values, the proposed weights have adjusted the regional average of precipitation. Figure 4 shows the temporal behavior of precipitation for each station, simple average (under equal weighting scheme), and regional average (under proposed weighting scheme). In contrast with equal weights, the proposed weighting scheme has diminished the impact of extreme values.

The following subsection presents the results associated with the estimation of BDWRDI and SPI.

4.2 Estimation of SPI, SPI (averaged), and BDWRDI under K-CKMD

This section presents some results related to the estimation of SPI and BDWRDI. Unlike traditional modeling, this research standardized the CDF of K-CGMM for SPI (based on individual station), SPI (Averaged), and BDWRDI. Figure 5 shows the density and scatter plots of the precipitation data of Astore and the whole regional data under the proposed weighting scheme. Here, the behavior of the precipitation data of other stations is similar to Astor. We assessed that regional data is more suitable for accurate inference compared to individual station modeling. For instance, the scatter plot under the proposed regional data is more accurate than the equally-weighted average. This is because the arithmetic means is greatly influenced by extreme value and outlier. Arithmetic means assigns equal weight to each observation. Consequently, the data having extreme values greatly influences the average. In contrast, the proposed weighting scheme reduces the impact of outliers and adjusts

Fig. 6 Time series behavior of SPI, SPI (Averaged) and BDWRDI at one-month time scale

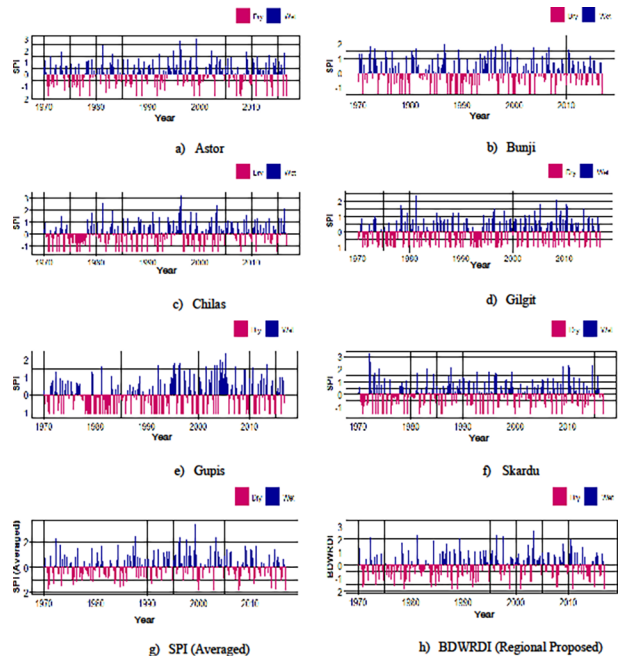


Table 3 Correlation of SPIs (of various stations) with SPI (Averaged) and BDWRDI

Time Scales	Index	Stations						
		Stat	Bunji	Chilas	Gilgit	Gupis	Skardu	
1	BDWRDI	r	0.8434	0.8563	0.8585	0.8367	0.6388	0.7909
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
	SPI (Averaged)	r	0.9088	0.8205	0.8411	0.7965	0.6355	0.8170
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
3	BDWRDI	r	0.8381	0.8655	0.8793	0.8479	0.6521	0.7526
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
	SPI (Averaged)	r	0.9112	0.8142	0.8752	0.7957	0.6416	0.8002
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
6	BDWRDI	r	0.8237	0.8379	0.8663	0.8423	0.6543	0.7132
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
	SPI (Averaged)	r	0.9060	0.7707	0.8499	0.7776	0.6485	0.7770
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
12	BDWRDI	r	0.7971	0.7897	0.8256	0.8208	0.6194	0.6869
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
	SPI (Averaged)	r	0.8615	0.7405	0.7791	0.7643	0.6626	0.7307
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
24	BDWRDI	r	0.7664	0.7590	0.7953	0.8068	0.6053	0.6393
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
	SPI (Averaged)	r	0.8275	0.6930	0.7471	0.7392	0.6620	0.6947
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
48	BDWRDI	r	0.6897	0.7116	0.7529	0.7147	0.6617	0.5301
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*
	SPI (Averaged)	r	0.7257	0.6331	0.7248	0.6176	0.7214	0.6079
		p-value	0.000*	0.000*	0.000*	0.000*	0.000*	0.000*

* at 0.05 level of significance

the data accordingly. Therefore, the BDWRDI drought indicator can potentially diminish the effects of extreme values and outliers. After fitting K-CGMM, the next step standardized the numerical CDF of K-CGMM. Figure 6 shows the temporal behavior of SPIs, SPI (averaged), and BDWRDI at a one-month time scale.

The next subsection assesses the suitability of BDWARDI using Pearson correlation.

4.3 Comparative analysis and validation

This subsection compares and validates the BDWRDI with individual station-based SPI and overall SPI using the Pearson correlation coefficient. Figure 6 shows consistency in the temporal behavior SPIs and BDWRDI at one a month time scale. In the remaining time scales,

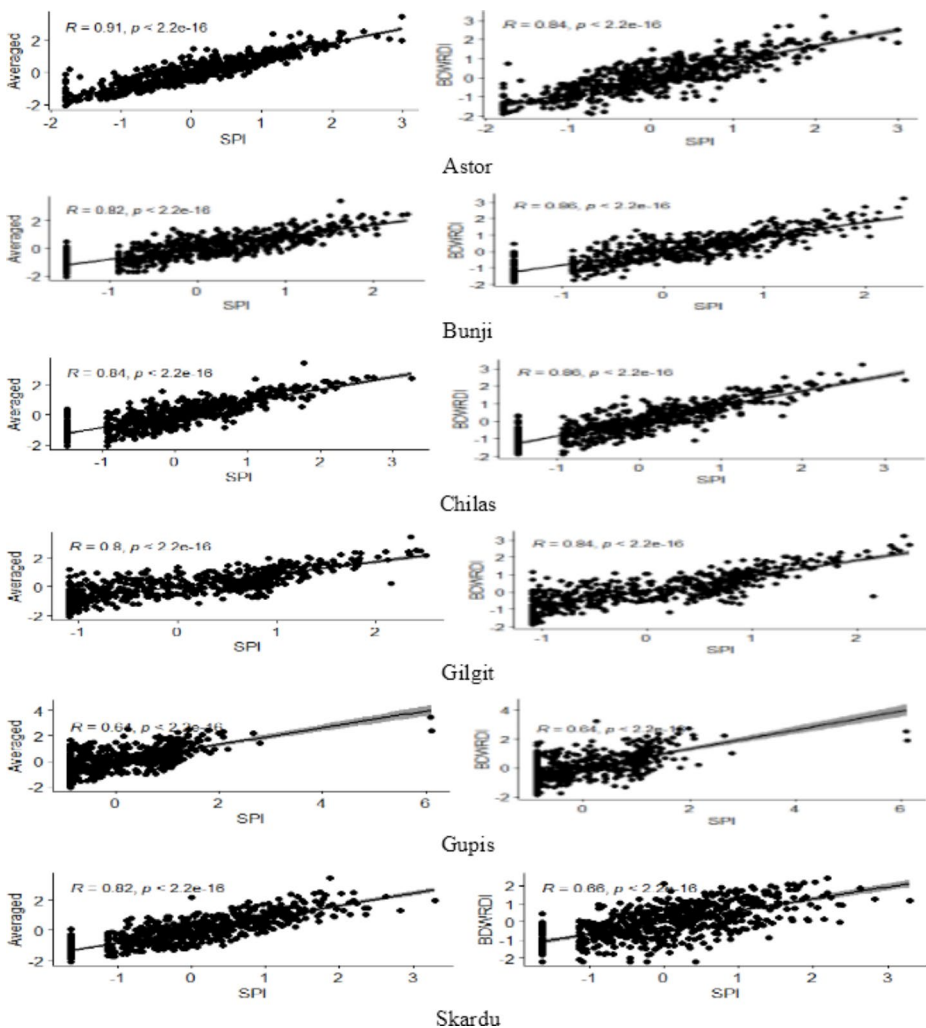


Fig. 7 Scatterplots and correlation of SPI with SPI (Averaged) and BDWRDI at one-month time scale

similar behavior has been assessed. Further, the study assesses linear behavior using scatter plots. Figure 7 reveals that the relation of BDWRDI with individual SPI is more linear than averaged SPI. Based on these graphical findings, it is rationally valid to compare BDWRDI with SPIs using Pearson correlation.

Table 3 provides the correlation values of average SPI and BDWRDI with individual SPIs. From correlation analysis, it is clearly evident that BDWRDI is homogeneously correlated with all SPIs. On the other hand, there is heterogeneous behavior of correlation among SPI (averaged) and SPIs. For instance, the BDWRDI has an approximately identical correlation with Astore (0.8434), Bunji (0.8563), Chilas (0.8585), and Gilgit (0.8367) at a one-month time scale. Contrariwise, SPI (averaged) has more variation in the correlation of all the selected stations. Similarly, results indicate that BDWRDI is more consistent with individual stations than SPI in other time scales. Further, p-values in all the correlation analyses are significant. These comparative findings prove that the proposed drought index (BDWRDI) is a good candidate drought index for regional drought monitoring.

5 Conclusions

Due to climate change and global warming, recurrent drought has badly affected many areas of life. The present paper developed a new state-of-the-art procedure for regional aggregation of precipitation in the presence of extreme values and outliers. The proposed procedure aggregates regional data of precipitation by diminishing the impact of outliers and extreme values. The application of the proposed method is based on six meteorological stations located in the Northern region of Pakistan. In comparison, the proposed indicator is compared with SPI using Pearson correlation. The numerical findings indicate that the proposed index has the potential to diminish the impact of outlier and extreme values. Relative to SPI, the proposed indicator is more flexible for characterizing regional climatology under global warming scenarios. These findings are helpful for drought practitioners to define regional climatology more accurately.

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Declarations

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References

- Abramowitz M, Stegun IA (eds) (1948) Handbook of mathematical functions with formulas, graphs, and mathematical tables, vol 55. US Government printing office
- Adedeji O, Olusola A, James G, Shaba HA, Orimoloye IR, Singh SK, Adelabu S (2020) Early warning systems development for agricultural drought assessment in Nigeria. *Environ Monit Assess* 192(12):1–21
- Ali F, Li BZ, Ali Z (2021b) Strengthening drought monitoring module by ensembling auxiliary information based varying estimators. *Water Resour Manage* 35(10):3235–3252
- Ali Z, Ellahi A, Hussain I, Nazeer A, Qamar S, Ni G, Faisal M (2021a) Reduction of Errors in Hydrological Drought Monitoring—A Novel Statistical Framework for Spatio-Temporal Assessment of Drought. *Water Resour Manage* 35(13):4363–4380
- Allan RP, Soden BJ (2008) Atmospheric warming and the amplification of precipitation extremes. *Science* 321(5895):1481–1484
- Arumugam P, Saranya R (2018) Outlier detection and missing value in seasonal ARIMA model using rainfall data. *Materials Today: Proceedings*, 5(1), 1791–1799
- Azadi H, Keramati P, Taheri F, Rafiaani P, Teklemariam D, Gebrehiwot K, Witlox F (2018) Agricultural land conversion: Reviewing drought impacts and coping strategies. *Int J disaster risk Reduct* 31:184–195
- Bahinipati CS (2020) Assessing the costs of droughts in rural India: a comparison of economic and non-economic loss and damage. *Curr Sci* 118(11):1832
- Balti H, Abbes AB, Mellouli N, Farah IR, Sang Y, Lamolle M (2020) A review of drought monitoring with big data: Issues, methods, challenges and research directions. *Ecol Inf* 60:101136
- Başakın EE, Ekmekcioğlu Ö (2020) Comment on “Comparison of the ability of ARIMA, WNN and SVM models for drought forecasting in the Sanjiang Plain, China” by Yuhu Zhang, Huirong Yang, Hengjian Cui, and Qiuhua Chen, in *Natural Resources Research*. *Nat Resour Res* 29(2):1465–1467
- Bong CHJ, Richard J (2020) Drought and climate change assessment using standardized precipitation index (SPI) for Sarawak River Basin. *J Water Clim Change* 11(4):956–965
- Byers EA, Coxon G, Freer J, Hall JW (2020) Drought and climate change impacts on cooling water shortages and electricity prices in Great Britain. *Nat Commun* 11(1):1–12
- Chevuturi A, Klingaman NP, Turner AG, Hannah S (2018) Projected changes in the Asian-Australian monsoon region in 1.5° C and 2.0° C global-warming scenarios. *Earth's Future* 6(3):339–358
- Delignette-Muller ML, Dutang C (2013) fitdistrplus: An R package for fitting distributions. *J Stat Softw* 64(4):1–34
- Gampe D, Zscheischler J, Reichstein M, O’Sullivan M, Smith WK, Sitch S, Buermann W (2021) Increasing impact of warm droughts on northern ecosystem productivity over recent decades. *Nat Clim Change* 11(9):772–779
- Giorgi F, Raffaele F, Coppola E (2019) The response of precipitation characteristics to global warming from climate projections. *Earth Sys Dyn* 10(1):73–89
- Gregersen IB, Sørup HJD, Madsen H, Rosbjerg D, Mikkelsen PS, Arnbjerg-Nielsen K (2013) Assessing future climatic changes of rainfall extremes at small spatio-temporal scales. *Clim Change* 118(3):783–797
- Hao Z, AghaKouchak A (2013) Multivariate standardized drought index: a parametric multi-index model. *Adv Water Resour* 57:12–18
- Huang J, Zhuo W, Li Y, Huang R, Sedano F, Su W, Zhang X (2020) Comparison of three remotely sensed drought indices for assessing the impact of drought on winter wheat yield. *Int J Digit Earth* 13(4):504–526
- Kao SC, Govindaraju RS (2010) A copula-based joint deficit index for droughts. *J Hydrol* 380(1–2):121–134
- Kamruzzaman M, Cho J, Jang MW, Hwang S (2019) Comparative evaluation of standardized precipitation index (SPI) and effective drought index (EDI) for meteorological drought detection over Bangladesh. *J Korean Soc Agricultural Eng* 61(1):145–159
- Khan MMH, Muhammad NS, El-Shafie A (2020) Wavelet based hybrid ANN-ARIMA models for meteorological drought forecasting. *J Hydrol* 590:125380
- Liu Y, Zhou R, Wen Z, Khalifa M, Zheng C, Ren H, Wang Z (2021) Assessing the impacts of drought on net primary productivity of global land biomes in different climate zones. *Ecol Ind* 130:108146
- Magalhães AR, Magalhães MC (2019) Drought preparedness and livelihood implications in developing countries: what are the options?—Latin America and Northeast Brazil. *Current directions in water scarcity research*, vol 2. Elsevier, pp 55–66
- McKee TB, Doesken NJ, Kleist J (1993), January The relationship of drought frequency and duration to time scales. In *Proceedings of the 8th Conference on Applied Climatology* (Vol. 17, No. 22, pp. 179–183)
- Mishra AK, Singh VP (2011) Drought modeling—A review. *J Hydrol* 403(1–2):157–175
- Nkemelang T, New M, Zaroug M (2018) Temperature and precipitation extremes under current, 1.5 C and 2.0 C global warming above pre-industrial levels over Botswana, and implications for climate change vulnerability. *Environ Res Lett* 13(6):065016

- Papalexiou SM, Montanari A (2019) Global and regional increase of precipitation extremes under global warming. *Water Resour Res* 55(6):4901–4914
- Tsakiris G, Pangalou D, Vangelis H (2007) Regional drought assessment based on the Reconnaissance Drought Index (RDI). *Water Resour Manage* 21(5):821–833
- Rasul G, Chaudhry QZ, Mahmood A, Hyder KW, Dahe Q (2011) Glaciers and glacial lakes under changing climate in Pakistan. *Pakistan Journal of Meteorology*, 8(15)
- Raza A, Hussain I, Ali Z, Faisal M, Elashkar EE, Shoukry AM, Gani S (2021) A seasonally blended and regionally integrated drought index using Bayesian network theory. *Meteorological Applications*, 28(3), e1992
- Salvador C, Nieto R, Linares C, Díaz J, Gimeno L (2020) Short-term effects of drought on daily mortality in Spain from 2000 to 2009. *Environ Res* 183:109200
- Spieß AN (2018) Propagation of Uncertainty Using Higher- Order Taylor Expansion and Monte Carlo Simulation
- Stagge JH, Tallaksen LM, Gudmundsson L, Van Loon AF, Stahl K (2015) Candidate distributions for climatological drought indices (SPI and SPEI). *Int J Climatol* 35(13):4027–4040
- Sugg M, Runkle J, Leeper R, Bagli H, Golden A, Handwerger LH, Woolard S (2020) A scoping review of drought impacts on health and society in North America. *Climatic Change*, 1–19
- Svoboda MD, Fuchs BA (2016) Handbook of drought indicators and indices. World Meteorological Organization, Geneva, Switzerland, pp 1–44
- Tabari H (2020) Climate change impact on flood and extreme precipitation increases with water availability. *Sci Rep* 10(1):1–10
- Vicente-Serrano SM, Beguería S, López-Moreno JI (2010) A multiscalar drought index sensitive to global warming: the standardized precipitation evapotranspiration index. *J Clim* 23(7):1696–1718
- Wei W, Zhang J, Zhou L, Xie B, Zhou J, Li C (2021) Comparative evaluation of drought indices for monitoring drought based on remote sensing data. *Environ Sci Pollut Res* 28(16):20408–20425
- Yaduvanshi A, Nkemelang T, Bendapudi R, New M (2021) Temperature and rainfall extremes change under current and future global warming levels across Indian climate zones. *Weather and Climate Extremes* 31:100291
- Zhang Y, Li Y, Ge J, Li G, Yu Z, Niu H (2018) Correlation analysis between drought indices and terrestrial water storage from 2002 to 2015 in China. *Environ Earth Sci* 77(12):1–12
- Zhang Y, Yang H, Cui H, Chen Q (2020) Comparison of the ability of ARIMA, WNN and SVM models for drought forecasting in the Sanjiang Plain, China. *Nat Resour Res* 29(2):1447–1464

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